

Question				Marking details	Marks available								
					AO1	AO2	AO3	Total	Maths	Prac			
2	(a)	(i)		{organic / nitrogenous} base and {pentose / 5C sugar} and {phosphate / PO ₄ ³⁻ / Pi }	1			1					
		(ii)		<table border="1"><tr><td>Structural Difference</td></tr><tr><td>ATP has 3 phosphate, RNA has one / ATP has more phosphates / Reference to different bases in ATP and RNA nucleotide or example of (1)</td></tr><tr><td>RNA has ribose, DNA has dexoxyribose / Less oxygen in DNA nucleotide or description of} / ORA / RNA has {uracil/H} and DNA has {thymine/CH₃} (1)</td></tr></table>	Structural Difference	ATP has 3 phosphate, RNA has one / ATP has more phosphates / Reference to different bases in ATP and RNA nucleotide or example of (1)	RNA has ribose, DNA has dexoxyribose / Less oxygen in DNA nucleotide or description of} / ORA / RNA has {uracil/H} and DNA has {thymine/CH ₃ } (1)	2			2		
Structural Difference													
ATP has 3 phosphate, RNA has one / ATP has more phosphates / Reference to different bases in ATP and RNA nucleotide or example of (1)													
RNA has ribose, DNA has dexoxyribose / Less oxygen in DNA nucleotide or description of} / ORA / RNA has {uracil/H} and DNA has {thymine/CH ₃ } (1)													
		(iii)		{Phosphate/ PO ₄ ³⁻ / Pi} and {nitrate/NO ₃ ⁻ }	1			1					
	(b)	(i)		76.9% (2) If incorrect award 1 mark for: 76.923 2 x10 ⁻¹¹ /2.6 x10 ⁻¹¹ x100 (1)		2		2	2				
		(ii)		Any one (x1) from Reference to: {protein synthesis / transcription / translation} is taking place / or description of (1) RNA is temporary (1) cell may have more or less ribosomes / OWTTE (1)		1		1					
	(c)			Energy is released when ATP is {hydrolysed / broken down} (to ADP and Pi) / or description of (1) {Energy is stored / ATP is made} when ADP and {phosphate / Pi} are joined / or description of / ADP + Pi → ATP (1) It is the source of energy in all {cells / reactions} in all organisms (1)	1	2		3					
				Question 2 total	5	5	0	10	2	0			